



Introduction to Linux and High Performance Computing

Gloria Nakabiri, David Twesigomwe, Ian Wasukira

What is Linux?

Linux is a free open-source derivative of the Unix Operating System (OS).

Unix

Original family of operating systems—often proprietary and commercially licensed

Distributions: IBM AIX, HP-UX, Sun Solaris



Linux

Open source OS based on Unix. It runs on computers, servers, mobile phones and a wide range of other devices

Distributions: Ubuntu, Redhat, Fedora, Debian, Android

Why Linux matters

01

OPEN

Free to use, modify
and distribute

02

STABLE

Built for reliable,
multi-user systems

03

FLEXIBLE

Runs across devices
and environments

04

ECOSYSTEM

Huge community +
scientific software

For researchers: reproducibility • automation • remote access • scale

Where is Linux used?

1

HPCs

2

WEB / CLOUD
SERVERS

3

PCs / DESKTOPS

4

SECURITY
SYSTEMS

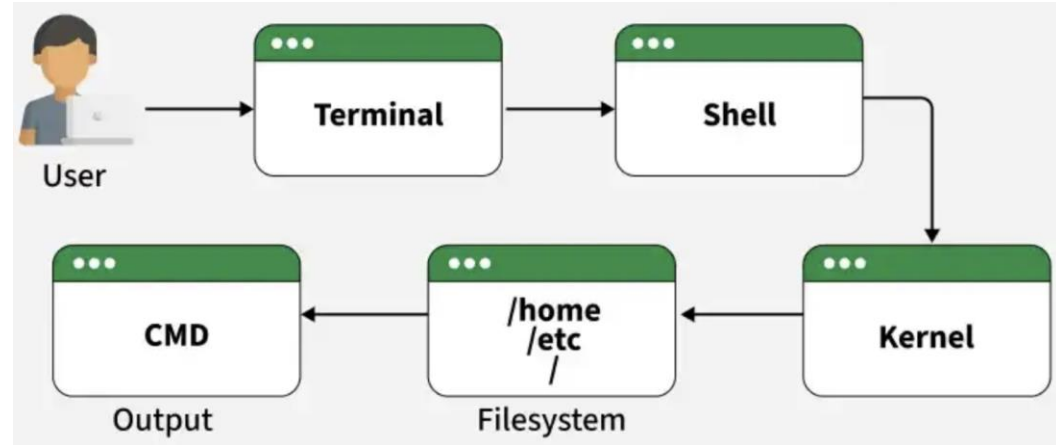
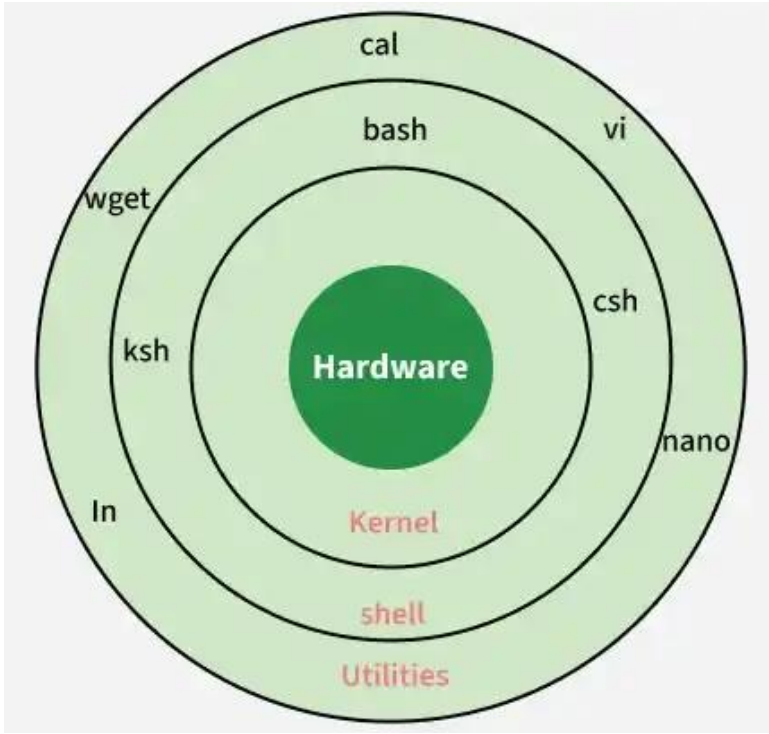
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PHONES

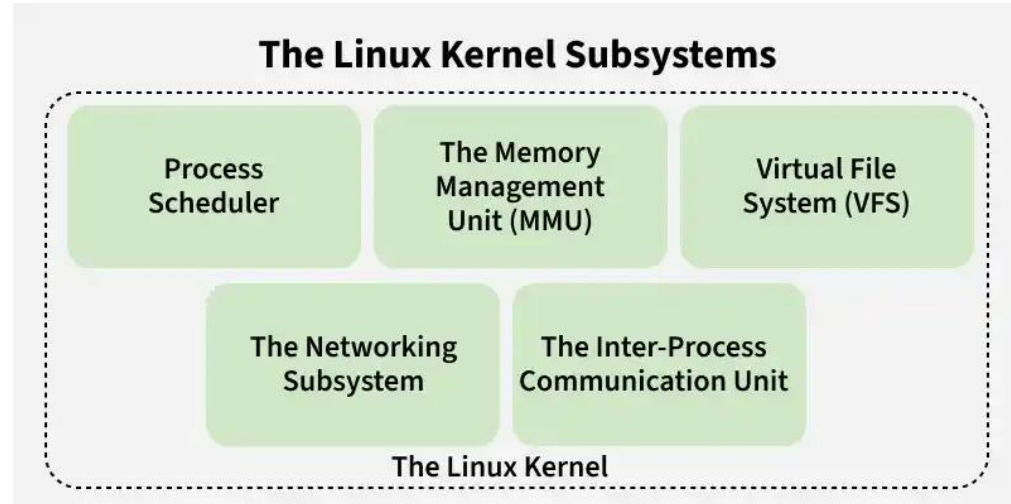
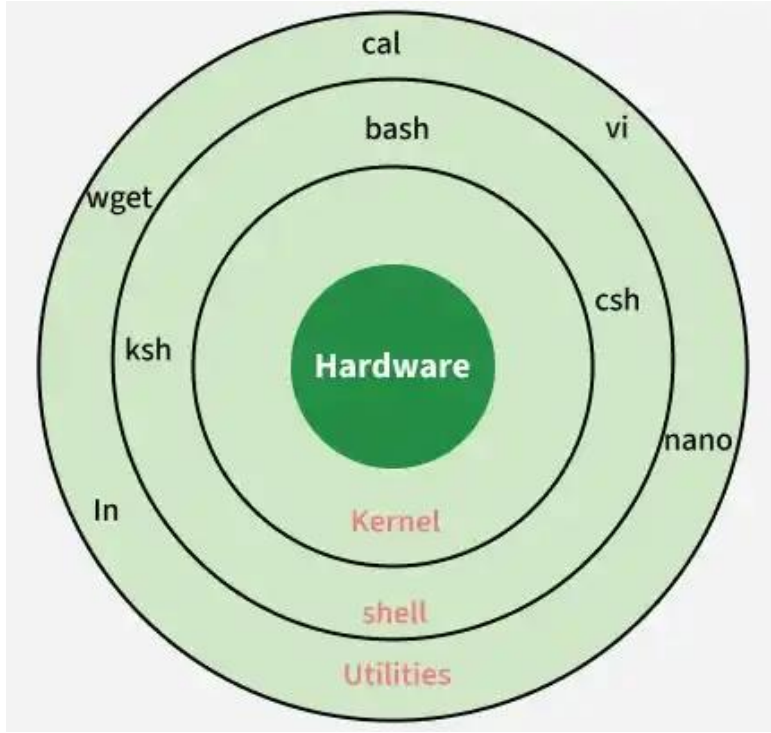
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OTHER DEVICES

Basic Linux Architecture

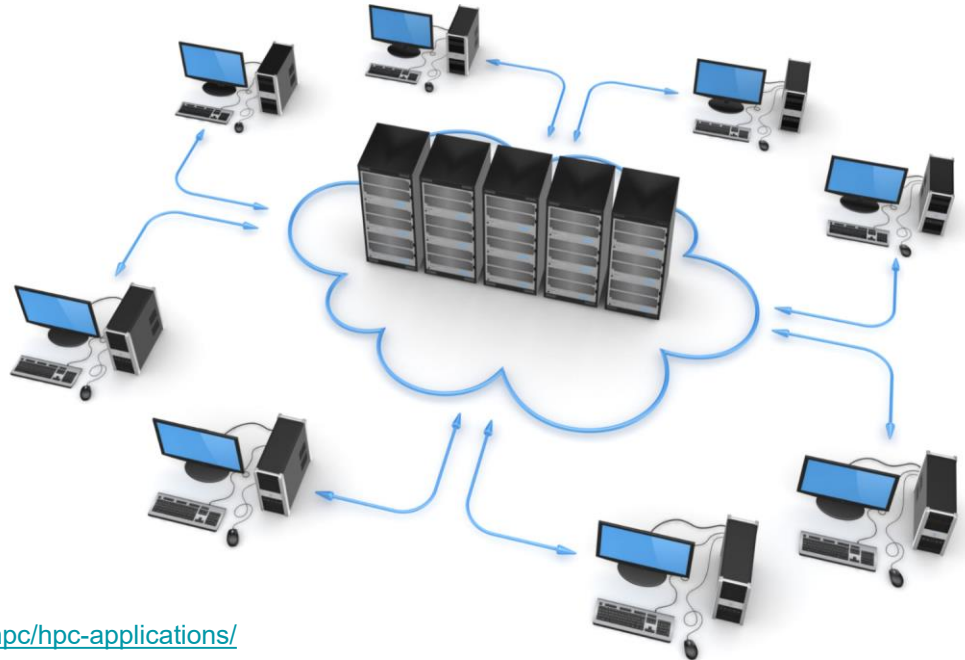


Basic Linux Architecture



What is high-performance computing (HPC)?

- Technology that uses clusters of powerful processors that work in parallel to process massive, multidimensional data sets and solve complex problems at extremely high speeds.



Why use HPC?

Advantages for data-intensive research



FASTER

Parallel processing reduces
time-to-result



BIGGER

Scale compute and storage
with growing datasets



RELIABILITY

Improved through efficient
data processing

The gain comes from matching the workload to the right resources

1 Genome Assembly

Assembling large genomes from millions to billions of reads.



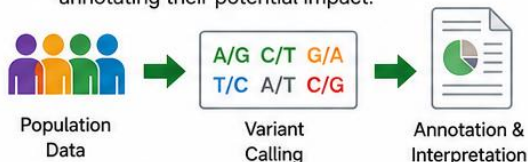
2 Read Alignment

Aligning vast numbers of reads to reference genomes.



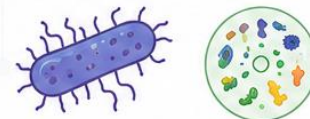
3 Variant Calling & Annotation

Detecting genetic variants and annotating their potential impact.



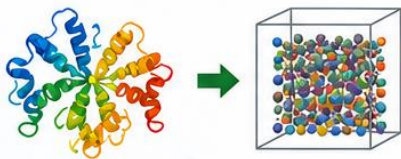
4 Pathogen Genomics & Microbiome Studies

Analyzing pathogens and complex microbial communities.



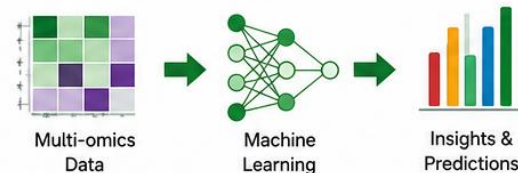
5 Protein Structure Prediction & Molecular Dynamics

Predicting 3D structures and simulating molecular interactions.



6 Machine Learning Integration

Using ML to find complex patterns in multi-omics datasets.



HPC
High Performance
Computing
for Bioinformatics

Challenges of HPC

1

COST

hardware + staff

2

SCALE

capacity planning

3

DATA

storage + networks

4

SKILLS

parallel thinking

5

SOFTWARE

compatibility

6

ENERGY

power + cooling

Good HPC practice is as much about resource management as raw computing power

Practical: HPC Orientation

01

Generate ssh key

```
ssh-keygen
```

02

Login to (ACE) HPC

```
ssh username@biocompace.ace.ac.ug
```

03

Where am I?

Key commands: `pwd` • `ls` • `cd`

04

What is available?

```
module avail
```

Practical: HPC Orientation

01

Load Conda

```
module load conda
```

02

Explore conda environments

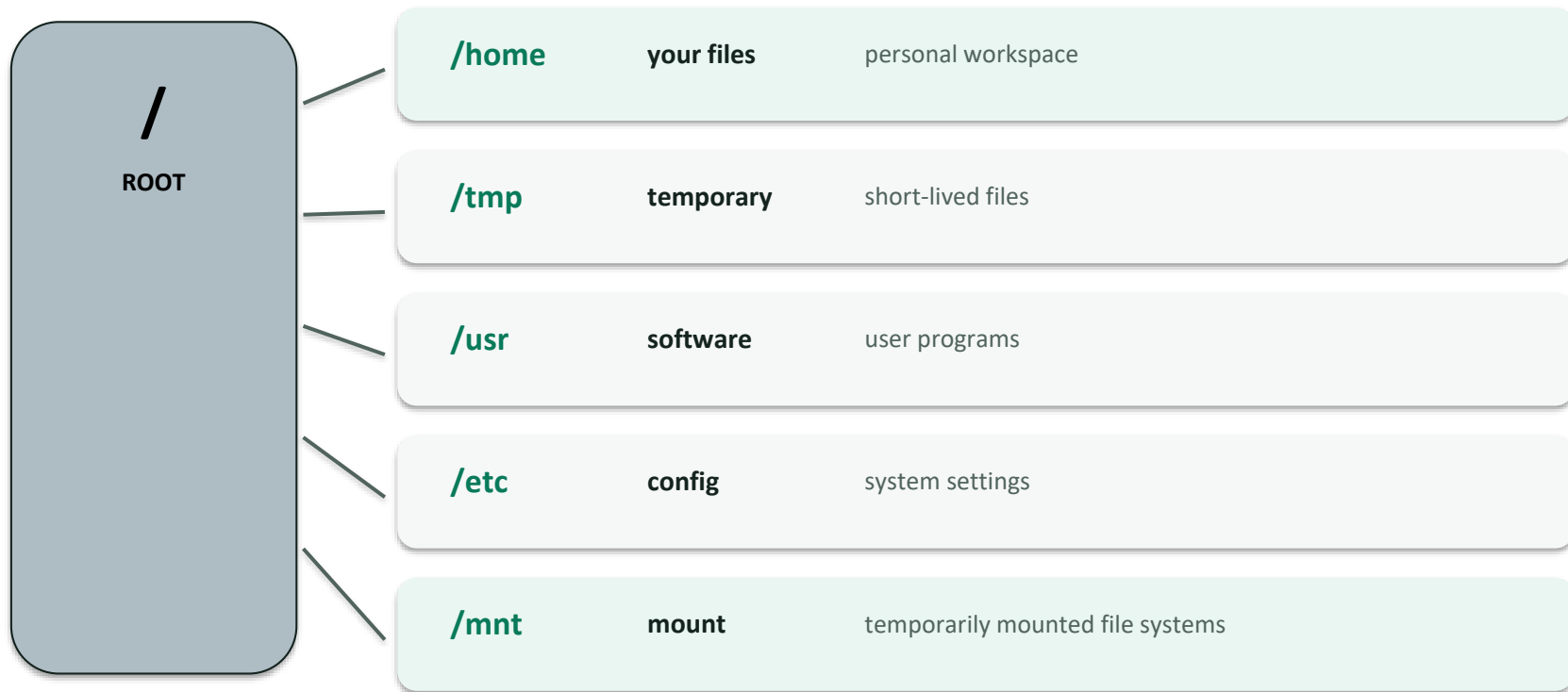
```
conda env list
```

03

Create new conda environment

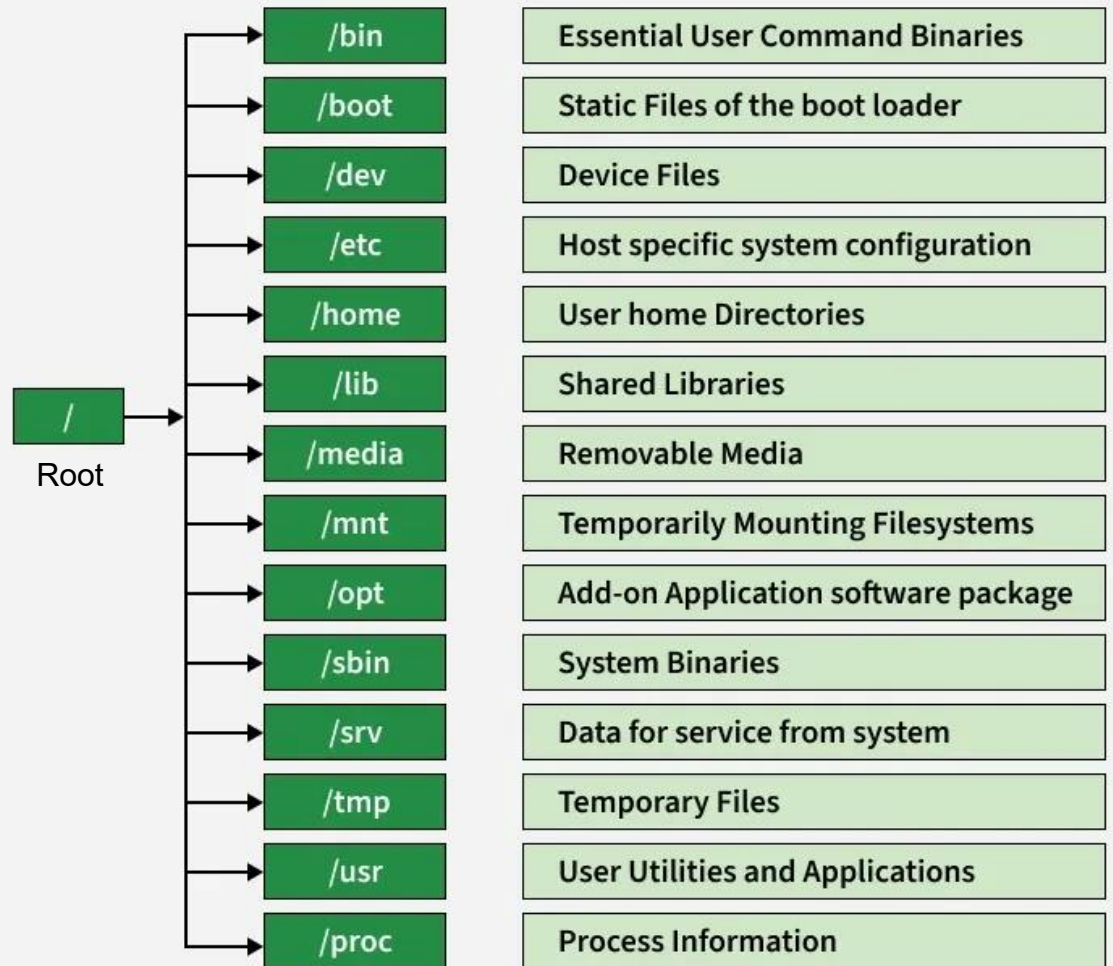
```
conda create --name myenv
```

Linux / HPC filesystem



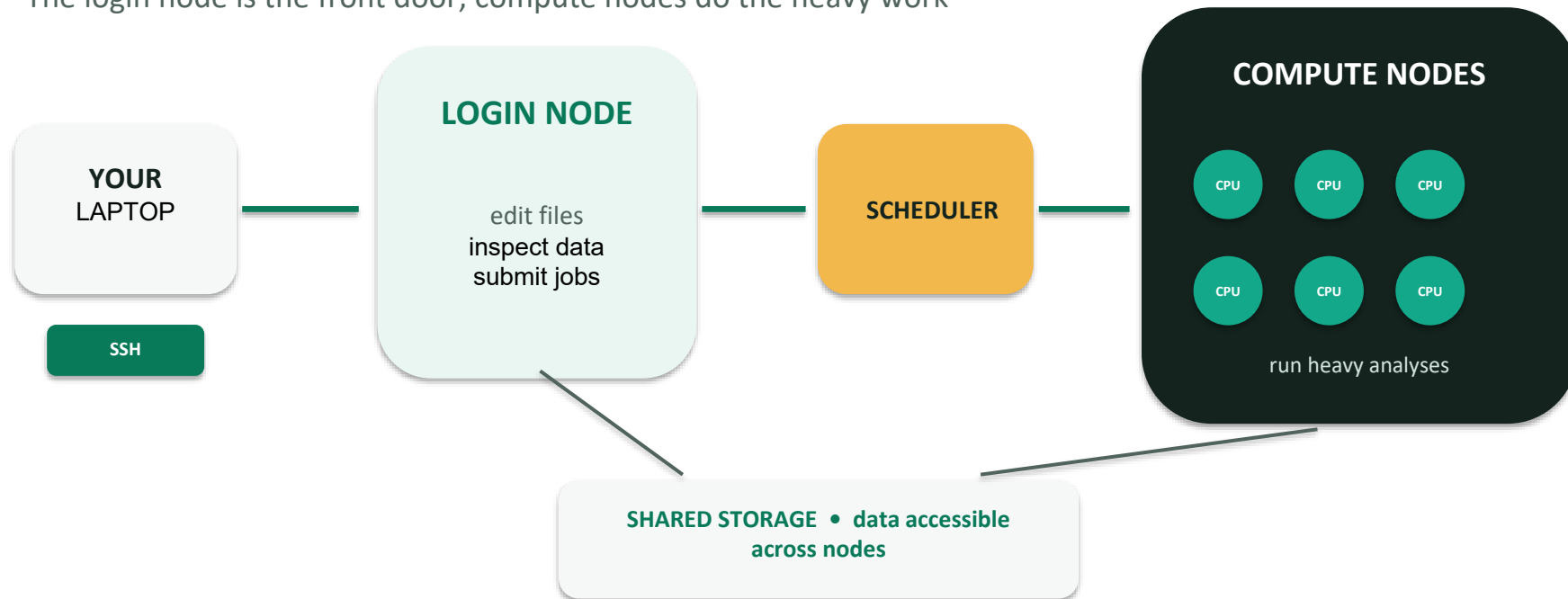
Exact shared/project paths vary by cluster — learn your local system.

Linux / HPC file structure



The HPC cluster: who does what?

The login node is the front door; compute nodes do the heavy work



Rule of thumb: do not run heavy analyses on the login node.

Thank you

```
$ echo "Questions?"  
> _
```